

Math211

Discrete Mathematics

Nested Quantifiers and Rules of Inference



Nested Quantifiers

- ▶ For every x, for every y: $\forall x \forall y P(x, y)$
 - ▶ Loop through x and for each x loop through y
 - ▶ If we find P(x, y) is true for all x and y, then the statement is true
 - ▶ If we ever hit a value x for which we hit a value for which P(x, y) is false, the whole statement is false
- ▶ For every x, there exists y: $\forall x \exists y P(x, y)$
 - ▶ Loop through x until we find a y that P(x, y) is true
 - ▶ If for every x, we find such a y (different), then the statement is true

Nested Quantifiers

- ▶ $\exists x \forall y P(x, y)$: loop through the values for x until we find an x for which $p(x, y)$ is always true when we loop through all values for y
 - ▶ Once found such one x , then it is true
- ▶ $\exists x \exists y P(x, y)$: loop through the values for x where for each x loop through the values of y until we find an x for which we find a y such that $p(x, y)$ is true
 - ▶ False only if we never hit an x for which we never find y such that $p(x, y)$ is true

Quantifications of Two Variables

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

Nested Quantifiers

Let $P(x, y)$ is $x+y = 0$. The **domain of x** is $\{-2, -1, 0\}$ and the **domain of y** is $\{1, 2, 0\}$. Determine whether each of the following statements is true or false. justify the false.

- | | | |
|----------------------------------|-------|---|
| 1. $\forall x \forall y P(x, y)$ | False | "For all x in $\{-2, -1, 0\}$ and for all y in $\{1, 2, 0\}$, $x+y=0$." |
| 2. $\forall y \forall x P(x, y)$ | False | "For all y in $\{1, 2, 0\}$, and for all x in $\{-2, -1, 0\}$, $x+y=0$." |
| 3. $\exists x \exists y P(x, y)$ | True | "There exists at least one x and one y such that $x+y=0$." |
| 4. $\exists y \exists x P(x, y)$ | True | "There exists at least one x and one y such that $x+y=0$." |

Nested Quantifiers

Let $P(x, y)$ is $x+y = 0$. The **domain of x** is $\{-2, -1, 0\}$ and the **domain of y** is $\{1, 2, 0\}$. Determine whether each of the following statements is true or false. justify the false.

- | | | |
|----------------------------------|-------|--|
| 1. $\forall x \exists y P(x, y)$ | True | "For every x , there exists some y (variant) such that $x+y=0$ " |
| 2. $\exists y \forall x P(x, y)$ | False | "There exists a y (unique) such that $x+y=0$ for all x " |
| 3. $\forall y \exists x P(x, y)$ | True | "For every y , there exists some x (variant) such that $x+y=0$ " |
| 4. $\exists x \forall y P(x, y)$ | False | "There exists a x (unique) such that $x+y=0$ for all y " |

Order of quantification

The order of quantifiers is important unless all the quantifiers are universal, or all the quantifiers are existential.

Quantification with more variables

Let $q(x, y, z)$ be the statement $x + y = z$,
and the domain is real number

$\forall x \forall y \exists z q(x, y, z)$: What does it mean? Is it true?

$\exists z \forall x \forall y q(x, y, z)$: What does it mean? Is it true?

The order of the
quantification
here is important

- ▶ $\forall x \forall y \exists z Q(x, y, z)$: “**For all real numbers x and for all real numbers y there is real number z (variant) such that $x + y = z$,**”. $\forall x \forall y \exists z Q(x, y, z)$: is **true**
- ▶ $\exists z \forall x \forall y Q(x, y, z)$: “**There is a real number z (constant) such that for all real numbers x and for all real numbers $x + y = z$,**”. $\forall x \forall y \exists z Q(x, y, z)$: is **false**, because there is no value of z that satisfies the equation $x + y = z$ for all values of x and y .

Translating mathematical statements

- ▶ “The sum of two positive integers is always positive”
- ▶ For every two integers (x and y), if these integers are both positive, then the sum of these integers is positive

$$\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x + y > 0))$$

where the domain for both variables consists of all integers

Domain: all integers ($\mathbb{Z}$)

Translating mathematical statements

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all integers

Domain: all integers (Z)

$$\forall x \forall y (x + y > 0)$$

where the domain for both variables consists of
all positive integers

Domain: all Positive integers

Example

- ▶ “Every real number except zero has a multiplicative inverse ($xy=1$)”
- ▶ “For every real number x except zero, x has a multiplicative inverse.
- ▶ “For every real number x , if $x \neq 0$, then there exists a real number y such that $xy = 1$.”

$$\forall x((x \neq 0) \rightarrow \exists y(xy = 1))$$

where the domain for both variables consists of real numbers

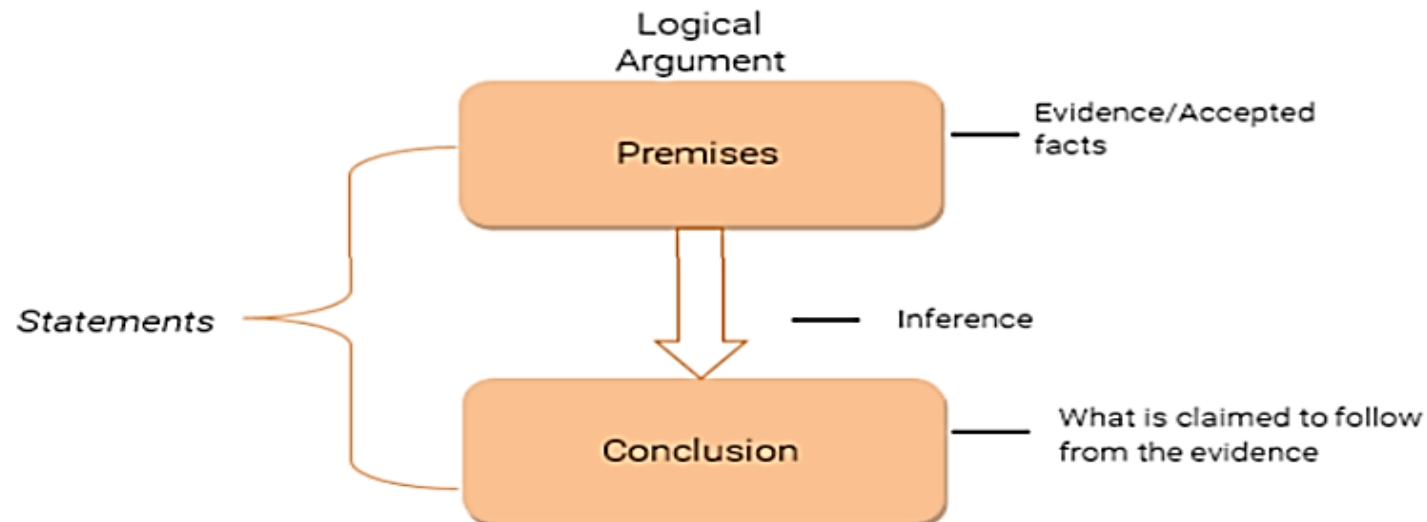
$$\forall x \exists y ((x \neq 0) \rightarrow (xy = 1))$$



1.6.1 Rules of Inference

Rules of Inference

- ▶ Inference is **deriving conclusions from evidences (premises)**.
- ▶ The rules of inference are the means used to draw conclusions from other assertions, and to derive an argument or a proof



Source: Patrick J. Hurley (2012)

Example

Consider **the following argument involving propositions** (which, by definition, is a sequence of propositions):

- ▶ “If you have a current password, then you can log onto the network.”
 - ▶ “You have a current password.”
- Premises**

Therefore,

- ▶ “You can log onto the network.”

Conclusion

Example

Use

- ▶ p to represent “You have a current password”
- ▶ q to represent “You can log onto the network”

Then, the argument has the form

$$\begin{array}{c} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

This argument is valid if $(p \rightarrow q) \wedge p \rightarrow q$ is a tautology.

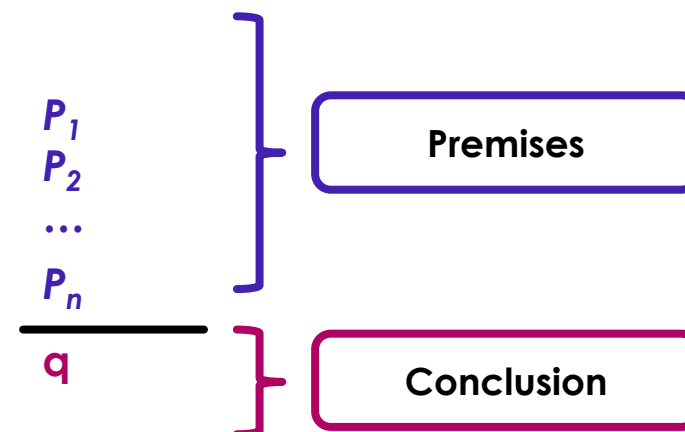
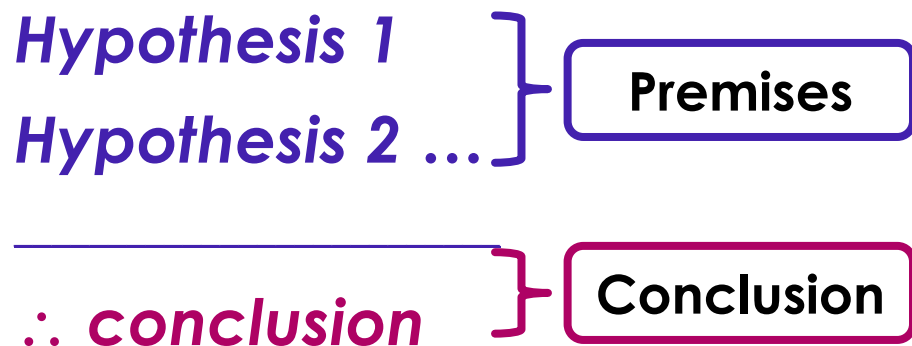
Example

p	q	$p \rightarrow q$	p	$(p \rightarrow q) \wedge p$	q	$((p \rightarrow q) \wedge p) \rightarrow q$
T	T	T	T	T	T	T
T	F	F	T	F	F	T
F	T	T	F	F	T	T
F	F	T	F	F	F	T

This argument is valid (can be used as an inference rule) if $(p \rightarrow q) \wedge p \rightarrow q$ is a tautology.

Valid Arguments in Propositional Logic

- ▶ An argument in propositional logic is a sequence of propositions. All the proposition in the argument are called premises and the final proposition is called the conclusion



This argument is valid (can be used as an inference rule) if $(P_1 \wedge P_2 \wedge \dots \wedge P_n) \rightarrow q$ is a tautology. Once the argument is valid, we can say that if P_1 is true and P_2 is true till P_n is true then q must be true.



Rules

Modus Ponens

$$\begin{array}{l} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

Rule of *modus ponens*
(a.k.a. *law of detachment*)

"the mode of
affirming"

- ▶ In logic terminology, modus ponens is the tautology:

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

- ▶ Example

- ▶ "if you are in this class, you will get a grade"
- ▶ "you are in this class"

Let p = "you are in this class"
Let q = "you will get a grade"

- ▶ By Modus Ponens, you can conclude that *you will get a grade*

Contrapositive

- ▶ The contrapositive is the following tautology

$$(p \rightarrow q) \equiv (\neg q \rightarrow \neg p)$$

- ▶ Usefulness
 - ▶ If you are having trouble proving the p implies q in a direct manner
 - ▶ You can try to prove the contrapositive instead!

Modus Tollens

$$\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$$

Rule of *modus tollens*

“the mode
of denying”

- ▶ Similar to the modus ponens, modus tollens is based on the following tautology

$$((p \rightarrow q) \wedge \neg q) \rightarrow \neg p$$

- ▶ Example

- ▶ “if you are in this class, you will get a grade”
- ▶ “you will not get a grade”

Let p = “you are in this class”
Let q = “you will get a grade”

- ▶ By Modus Tollens, you can conclude that *you are not in this class*

Addition

$$\frac{p}{\therefore p \vee q}$$

Rule of Addition

- ▶ Addition involves the tautology

$$p \rightarrow (p \vee q)$$

- ▶ Example: “It is below freezing now. Therefore, it is either below freezing or raining now.”

Simplification

$$\frac{p \wedge q}{\therefore p}$$

Rule of Simplification

- ▶ Simplification is based on the tautology

$$(p \wedge q) \rightarrow p$$

$$(p \wedge q) \rightarrow q$$

- ▶ *Example: “It is below freezing and raining now. Therefore, it is below freezing now.”*

Conjunction

$$\frac{p \quad q}{\therefore p \wedge q}$$

Rule of Conjunction

- ▶ The conjunction is almost trivially intuitive.
- ▶ It is based on the following tautology:

$$((p) \wedge (q)) \rightarrow (p \wedge q)$$

Hypothetical Syllogism

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Rule of hypothetical syllogism

- ▶ Hypothetical syllogism is based on the following tautology
$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$
- ▶ Example:
 - ▶ If you don't get a job, you won't have money
 - ▶ If you don't have money, you will starve.
 - ▶ Therefore, if you don't get a job, you'll starve

Disjunctive Syllogism

$$\begin{array}{c} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

Rule of disjunctive syllogism

- ▶ A disjunctive syllogism is formed on the basis of the tautology

$$((p \vee q) \wedge \neg p) \rightarrow q$$

- ▶ Example

- ▶ The sky is either blue or grey
- ▶ Well it isn't blue
- ▶ Therefore, the sky is grey

Resolution

$$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$$

Resolution

- ▶ For resolution, we have the following tautology

$$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$$

- ▶ Essentially,
 - ▶ If we have two true disjunctions that have mutually exclusive propositions
 - ▶ Then we can conclude that the disjunction of the two non-mutually exclusive propositions is true

Rules of Inference

<i>Rule of Inference</i>	<i>Tautology</i>	<i>Name</i>
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism

Rules of Inference

$\frac{p}{\therefore p \vee q}$	$p \rightarrow (p \vee q)$	Addition
$\frac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Simplification
$\frac{p}{\therefore p \wedge q}$ q	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\frac{p \vee q}{\therefore q \vee r}$ $\neg p \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Example 1

Premise: $p \wedge (p \rightarrow q)$

Conclusion: q

1	$p \wedge (p \rightarrow q)$	Premise	
2	p	Simplification on 1	$\therefore (p \wedge q)$ $\therefore p$
3	$p \rightarrow q$	Simplification on 1	
4	q	Modus Ponens on 2 & 3	$\therefore [(p \rightarrow q) \wedge p]$ $\therefore q$

$p \rightarrow q$ $\underline{p}$ $\therefore q$	<i>modus ponens</i>
$p \rightarrow q$ $\underline{\neg q}$ $\therefore \neg p$	<i>modus tollens</i>
$\underline{p}$ $\therefore p \vee q$	Addition
$\underline{p \wedge q}$ $\therefore p$	Simplification
$\underline{p}$ $\underline{q}$ $\therefore p \wedge q$	Conjunction
$p \rightarrow q$ $\underline{q \rightarrow r}$ $\therefore p \rightarrow r$	hypothetical syllogism
$p \vee q$ $\underline{\neg p}$ $\therefore q$	disjunctive syllogism
$p \vee q$ $\underline{\neg p \vee r}$ $\therefore q \vee r$	Resolution

Example 2

- ▶ Assume that the statements below hold:
 - $(p \rightarrow q)$
 - $(r \rightarrow s)$
 - $(r \vee p)$
- ▶ Assume that q is false
- ▶ Show that s must be true

Premises

$(p \rightarrow q)$
 $(r \rightarrow s)$
 $(r \vee p)$
 $\sim q$

Conclusion

s

$$\begin{array}{l}
 p \rightarrow q \\
 r \rightarrow s \\
 r \vee p \\
 \neg q \\
 \hline
 s
 \end{array}$$

1. $(p \rightarrow q)$

Premise

2. $\neg q$

Premise

3. $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$

by modus tollens on 1 + 2

4. $(r \vee p) = (p \vee r)$

Premise

5. $(p \vee r) \wedge \neg p \rightarrow r$

by disjunctive syllogism 3 + 4

6. $(r \rightarrow s)$

Premise

7. $(r \wedge (r \rightarrow s)) \rightarrow s$

by modus ponens 5 + 6

$p \rightarrow q$ $\underline{p}$ $\therefore q$	modus ponens
$p \rightarrow q$ $\underline{\neg q}$ $\therefore \neg p$	modus tollens
$\underline{p}$ $\therefore p \vee q$	Addition
$\underline{p \wedge q}$ $\therefore p$	Simplification
p $\underline{q}$ $\therefore p \wedge q$	Conjunction
$p \rightarrow q$ $\underline{q \rightarrow r}$ $\therefore p \rightarrow r$	hypothetical syllogism
$p \vee q$ $\underline{\neg p}$ $\therefore q$	disjunctive syllogism
$p \vee q$ $\underline{\neg p \vee r}$ $\therefore q \vee r$	Resolution

Example 3

- ▶ Suppose we have the following premises:
 - “It is not sunny, and it is cold.”**
 - “If it is not sunny, we will not swim.”**
 - “If we do not swim, then we will go boating.”**
 - “If we go boating, then we will be home early.”**
- ▶ Given these premises, prove the theorem
“We will be home early” using inference rules.

Example 3

“It is not sunny, and it is cold.”

“if it is not sunny, we will not swim”

“If we do not swim, then we will go boating.”

“If we go boating, then we will be home early.”

- ▶ Given these premises, prove the theorem
“We will be home early” using inference rules.

Let us adopt the following abbreviations:

sunny = **“It is sunny”**; *cold* = **“It is cold”**;
swim = **“We will swim”**; *boating* = **“We will go boating”**;

early = **“We will be home early”**.

Example 3

“It is not sunny, and it is cold.”

“if it is not sunny, we will not swim”

“If we do not swim, then we will go boating.”

“If we go boating, then we will be home early.”

- ▶ Given these premises, prove the theorem
“We will be home early” using inference rules.

Let us adopt the following abbreviations:

sunny = **“It is sunny”**; *cold* = **“It is cold”**;
swim = **“We will swim”**; *boating* = **“We will go boating”**;

early = **“We will be home early”**.

Then, the premises can be written as:

- (1) $\neg \text{sunny} \wedge \text{cold}$
- (2) $\neg \text{sunny} \rightarrow \neg \text{swim}$
- (3) $\neg \text{swim} \rightarrow \text{boating}$
- (4) $\text{boating} \rightarrow \text{early}$

$\neg \text{sunny} \wedge \text{cold}$
 $\neg \text{sunny} \rightarrow \neg \text{swim}$
 $\neg \text{swim} \rightarrow \text{boating}$
 $\text{boating} \rightarrow \text{early}$

 early

Steps

1. $\neg \text{sunny} \wedge \text{cold}$
2. $\neg \text{sunny}$
3. $\neg \text{sunny} \rightarrow \neg \text{swim}$
4. $\neg \text{swim}$
5. $\neg \text{swim} \rightarrow \text{boating}$
6. boating
7. $\text{boating} \rightarrow \text{early}$
8. early

Proved by

- Premise #1.
Simplification of 1.
Premise #2.
Modus ponens on 2,3.
Premise #3.
Modus ponens on 4,5.
Premise #4.
Modus ponens on 6,7.

$\frac{p \rightarrow q \quad p}{\therefore q}$	modus ponens
$\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$	modus tollens
$\frac{p}{\therefore p \vee q}$	Addition
$\frac{p \wedge q}{\therefore p}$	Simplification
$\frac{p \quad q}{\therefore p \wedge q}$	Conjunction
$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	hypothetical syllogism
$\frac{p \vee q \quad \neg p}{\therefore q}$	disjunctive syllogism
$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$	Resolution

Inference Rules for Quantifiers

Rules of inference can be extended in a straightforward manner to quantified statements

1. **Universal Instantiation (UI)**
2. **Universal Generalization (UG)**
3. **Existential Instantiation (EI)**
4. **Existential Generalization (EG)**

Universal Instantiation (UI)

$$\frac{\forall x P(x)}{\therefore P(o)} \quad (\text{substitute } x \text{ with any object } o \text{ in the domain})$$

Example

- ▶ All dogs are cute
- ▶ Oliver is a dog
- ▶ Therefore, Oliver is cute

$$\text{UI} = \frac{\forall x P(x)}{\therefore P(c)}$$

Universal Generalization (UG)

$$\text{UG} = \frac{P(c)}{\therefore \forall x P(x)} \quad (\text{for } c \text{ a general element of u.d.})$$

We will mainly focus on valid cases

Example 1:

- ▶ Oliver is a dog that has 4 legs.
- ▶ Then, All dogs have 4 legs.

Valid

Example 2:

- ▶ Ahmed is an Egyptian that loves falafel.
- ▶ Then, ALL Egyptians love falafel

Invalid

Inference Rules for Quantifiers

Existential Instantiation (EI)

$$\frac{\exists x P(x)}{\therefore P(c)} \quad (\text{substitute a new constant } c)$$

Existential Generalization (EG)

$$\frac{P(o)}{\therefore \exists x P(x)} \quad (\text{substitute any extant object } o)$$

Rules of Inference for Quantified Statements

<i>Rule of Inference</i>	<i>Name</i>
$\frac{\forall x P(x)}{\therefore P(c)}$	Universal instantiation
$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$	Universal generalization
$\frac{\exists x P(x)}{\therefore P(c) \text{ for some element } c}$	Existential instantiation
$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$	Existential generalization

Example 1

- ▶ Show that “A car in the garage has an engine problem” and “Every car in the garage has been sold” imply the conclusion “A car has been sold has an engine problem”
- ▶ Let
 - ▶ $G(x)$: “x is in the garage”
 - ▶ $E(x)$: “x has an engine problem”
 - ▶ $S(x)$: “x has been sold”
- ▶ Let the domain be the set of all cars
- ▶ The premises are as follows:
 - ▶ $\exists x (G(x) \wedge E(x))$
 - ▶ $\forall x (G(x) \rightarrow S(x))$
- ▶ The conclusion we want to show is: $\exists x (S(x) \wedge E(x))$

$$\begin{array}{l} \exists x (G(x) \wedge E(x)) \\ \forall x (G(x) \rightarrow S(x)) \end{array}$$

$$\exists x (S(x) \wedge E(x))$$

$\frac{\forall x P(x)}{\therefore P(c)}$	Universal Instantiation	$\frac{p \rightarrow q}{p} \therefore q$	<i>modus ponens</i>
$\frac{P(c)}{\therefore \forall x P(x)}$	Universal Generalization	$\frac{p \rightarrow q}{\neg q} \therefore \neg p$	<i>modus tollens</i>
$\frac{\exists x P(x)}{\therefore P(c)}$	Existential Instantiation	$\frac{p}{\therefore p \vee q}$	Addition
$\frac{P(c)}{\therefore \exists x P(x)}$	Existential Generalization	$\frac{p \wedge q}{\therefore p}$	Simplification
		$\frac{p}{q} \therefore p \wedge q$	Conjunction
		$\frac{p \rightarrow q}{q \rightarrow r} \therefore p \rightarrow r$	hypothetical syllogism
		$\frac{p \vee q}{\neg p} \therefore q$	disjunctive syllogism
		$\frac{p \vee q}{\neg p \vee r} \therefore q \vee r$	Resolution

- | | | |
|----|-------------------------------------|--|
| 1. | $\exists x (G(x) \wedge E(x))$ | <i>1st premise</i> |
| 2. | $(G(c) \wedge E(c))$ | <i>Existential instantiation of (1)</i> |
| 3. | $G(c)$ | <i>Simplification of (2)</i> |
| 4. | $\forall x (G(x) \rightarrow S(x))$ | <i>2nd premise</i> |
| 5. | $G(c) \rightarrow S(c)$ | <i>Universal instantiation of (4)</i> |
| 6. | $S(c)$ | <i>Modus ponens on (3) and (5)</i> |
| 7. | $E(c)$ | <i>Simplification from (2)</i> |
| 8. | $S(c) \wedge E(c)$ | <i>Conjunction of (6) and (7)</i> |
| 9. | $\exists x (S(x) \wedge E(x))$ | <i>Existential generalization of (8)</i> |

Example 2

“Everyone in this discrete math class has taken a course in computer science” and “Ali is a student in this class” imply “Ali has taken a course in computer science”. Assume the domain consists of all students.

$D(x)$: “ x is in discrete math class”

$C(x)$: “ x has taken a course in computer science”

$$\forall x (D(x) \rightarrow C(x))$$

$$D(\text{Ali})$$

$$\therefore C(\text{Ali})$$

$$\forall x (D(x) \rightarrow C(x))$$

$$D(\text{Ali})$$

$$\therefore C(\text{Ali})$$

$\frac{\forall x P(x)}{\therefore P(c)}$	Universal Instantiation
$\frac{P(c)}{\therefore \forall x P(x)}$	Universal Generalization
$\frac{\exists x P(x)}{\therefore P(c)}$	Existential Instantiation
$\frac{P(c)}{\therefore \exists x P(x)}$	Existential Generalization

$\frac{p \rightarrow q \quad p}{\therefore q}$	<i>modus ponens</i>
$\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$	<i>modus tollens</i>
$\frac{p}{\therefore p \vee q}$	Addition
$\frac{p \wedge q}{\therefore p}$	Simplification
$\frac{p \quad q}{\therefore p \wedge q}$	Conjunction
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$\frac{p \vee q \quad \neg p}{\therefore q}$	disjunctive syllogism
$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$	Resolution

Step

Proved by

1. $\forall x (D(x) \rightarrow C(x))$
2. $D(\text{Ali}) \rightarrow C(\text{Ali})$
3. $D(\text{Ali})$
4. $C(\text{Ali})$

Premise #1.
Univ. instantiation.
Premise #2.
Modus ponens on 2,3.

Example 3

“A student in this class has not read the book” and “Everyone in this class passed the first exam” imply “Someone who passed the first exam has not read the book”. Assume the domain consists of all student.

$C(x)$: “ x is in this class”

$R(x)$: “ x has read the book”

$P(x)$: “ x passed the first exam”

$$\exists x (C(x) \wedge \neg R(x))$$

$$\forall x (C(x) \rightarrow P(x))$$

$$\therefore \exists x (P(x) \wedge \neg R(x))$$

$$\exists x (C(x) \wedge \neg R(x))$$

$$\forall x (C(x) \rightarrow P(x))$$

$$\therefore \exists x (P(x) \wedge \neg R(x))$$

$$\frac{\forall x P(x)}{\therefore P(c)}$$

Universal
Instantiation

$$\frac{P(c)}{\therefore \forall x P(x)}$$

Universal
Generalization

$$\frac{\exists x P(x)}{\therefore P(c)}$$

Existential
Instantiation

$$\frac{P(c)}{\therefore \exists x P(x)}$$

Existential
Generalization

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

*modus
ponens*

$$\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$$

modus tollens

$$\frac{p}{\therefore p \vee q}$$

Addition

$$\frac{p \wedge q}{\therefore p}$$

Simplification

$$\frac{p \quad q}{\therefore p \wedge q}$$

Conjunction

$$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$$

hypothetical
syllogism

$$\frac{p \vee q \quad \neg p}{\therefore q}$$

disjunctive
syllogism

$$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$$

Resolution

Step

1. $\exists x (C(x) \wedge \neg R(x))$

2. $C(a) \wedge \neg R(a)$

3. $C(a)$

4. $\forall x (C(x) \rightarrow P(x))$

5. $C(a) \rightarrow P(a)$

6. $P(a)$

7. $\neg R(a)$

8. $P(a) \wedge \neg R(a)$

9. $\exists x (P(x) \wedge \neg R(x))$

Proved by

Premise #1.

Exist. Instantiation of 1.

Simplification on 2.

Premise #2.

Univ. instantiation of 4.

Modus ponens on 3,5

Simplification on 2

Conjunction on 6,7

Exist. Generalization of 8

Problem 1

The premises

- ▶ “If you send me an e-mail message, then I will finish writing the program”
- ▶ “If you do not send me an e-mail message, then I will go to sleep early”
- ▶ “If I go to sleep early, then I will wake up feeling refreshed”

The conclusion

- ▶ “If I do not finish writing the program, then I will wake up feeling refreshed.”

Solution

- ▶ p be the proposition “You send me an e-mail message,”
 - ▶ q the proposition “I will finish writing the program”
 - ▶ r the proposition “I will go to sleep early”
 - ▶ s the proposition “I will wake up feeling refreshed”
-
- ▶ Then the premises are $p \rightarrow q$, $\neg p \rightarrow r$, and $r \rightarrow s$
 - ▶ The desired conclusion is $\neg q \rightarrow s$

$$\begin{array}{l}
 p \rightarrow q \\
 \neg p \rightarrow r \\
 r \rightarrow s
 \end{array}$$

$$\therefore \neg q \rightarrow s$$

Step	Reason
1. $p \rightarrow q$	Premise
2. $\neg q \rightarrow \neg p$	Contrapositive of (1)
3. $\neg p \rightarrow r$	Premise
4. $\neg q \rightarrow r$	Hypothetical syllogism using (2) and (3)
5. $r \rightarrow s$	Premise
6. $\neg q \rightarrow s$	Hypothetical syllogism using (4) and (5)

$ \begin{array}{l} p \rightarrow q \\ \underline{p} \\ \therefore q \end{array} $	<i>modus ponens</i>
$ \begin{array}{l} p \rightarrow q \\ \underline{\neg q} \\ \therefore \neg p \end{array} $	<i>modus tollens</i>
$ \begin{array}{l} \underline{p} \\ \therefore p \vee q \end{array} $	Addition
$ \begin{array}{l} \underline{p \wedge q} \\ \therefore p \end{array} $	Simplification
$ \begin{array}{l} p \\ \underline{q} \\ \therefore p \wedge q \end{array} $	Conjunction
$ \begin{array}{l} p \rightarrow q \\ \underline{q \rightarrow r} \\ \therefore p \rightarrow r \end{array} $	hypothetical syllogism
$ \begin{array}{l} p \vee q \\ \underline{\neg p} \\ \therefore q \end{array} $	disjunctive syllogism
$ \begin{array}{l} p \vee q \\ \underline{\neg p \vee r} \\ \therefore q \vee r \end{array} $	Resolution

Problem 2

- ▶ Is this argument correct or incorrect?
 - ▶ “**All** TAs compose easy quizzes. Rana is a TA. Therefore, Rana composes easy quizzes.”
 - ▶ Assume the domain is all people

Solution

- ▶ First, separate the premises from conclusions:
 - ▶ Premise #1: All TAs compose easy quizzes.
 - ▶ Premise #2: Rana is a TA.
 - ▶ Conclusion: Rana composes easy quizzes.

Next, re-render the example in logic notation.

- ▶ Premise #1: All TAs compose easy quizzes.
 - ▶ Let U.D. = all people
 - ▶ Let $T(x) \equiv$ “x is a TA”
 - ▶ Let $E(x) \equiv$ “x composes easy quizzes”
 - ▶ Then Premise #1 says: $\forall x, T(x) \rightarrow E(x)$

Solution cont...

- ▶ Premise #2: Rana is a TA.
 - ▶ Let $R \equiv \text{Rana}$
 - ▶ Then Premise #2 says: $T(\text{Rana})$
- ▶ Conclusion says: $E(\text{Rana})$
- ▶ The argument is correct, because it can be reduced to a sequence of applications of valid inference rules, as follows:

$$\forall x T(x) \rightarrow E(x)$$

$$T(\text{Rana})$$

$$\therefore E(\text{Rana})$$

$\frac{\forall x P(x)}{\therefore P(c)}$	Universal Instantiation
$\frac{P(c)}{\therefore \forall x P(x)}$	Universal Generalization
$\frac{\exists x P(x)}{\therefore P(c)}$	Existential Instantiation
$\frac{P(c)}{\therefore \exists x P(x)}$	Existential Generalization

$\frac{p \rightarrow q \quad p}{\therefore q}$	<i>modus ponens</i>
$\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$	<i>modus tollens</i>
$\frac{p}{\therefore p \vee q}$	Addition
$\frac{p \wedge q}{\therefore p}$	Simplification
$\frac{p \quad q}{\therefore p \wedge q}$	Conjunction
$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	hypothetical syllogism
$\frac{p \vee q \quad \neg p}{\therefore q}$	disjunctive syllogism
$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$	Resolution

►	<u>Statement</u>	<u>How obtained</u>
1.	$\forall x, T(x) \rightarrow E(x)$	(Premise #1)
2.	$T(\text{Rana}) \rightarrow E(\text{Rana})$	(Universal instantiation)
3.	$T(\text{Rana})$	(Premise #2)
4.	$E(\text{Rana})$	(Modus Ponens from statements #2 and #3)

Supplementary Material

- ▶ Discrete Math - 1.6.1 Rules of Inference for Propositional Logic
- ▶ Discrete Math - 1.6.2 Rules of Inference for Quantified Statements
- ▶ When to use “Conjunction” & when to use “Implies”

Next Lecture

- ▶ Introduction to proofs
 - ▶ direct proof, proof by contraposition, proof by contradiction, proof by cases.

See you next lecture

